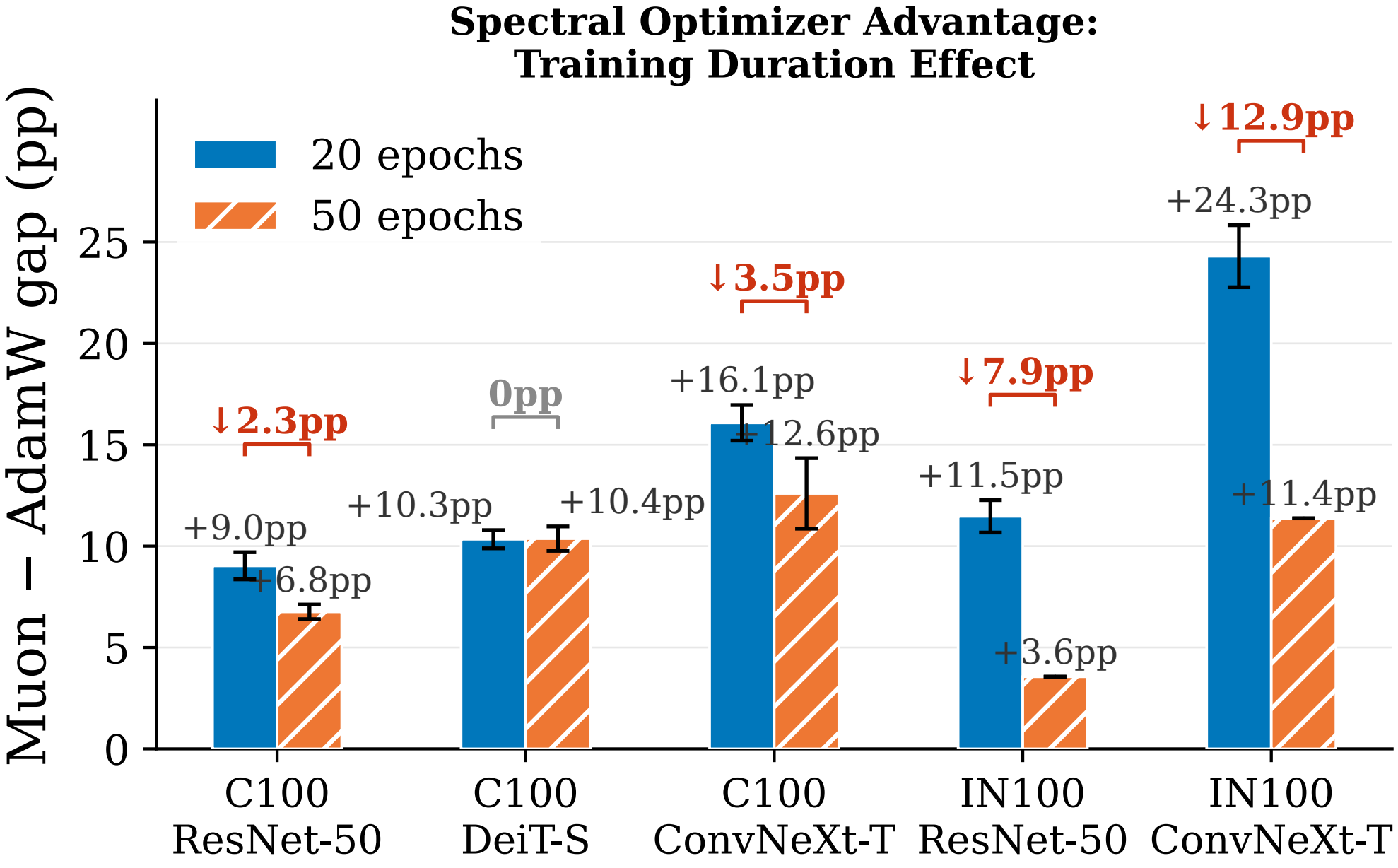
Spectral Optimizer Advantage:
Training Duration Effect
Muon − AdamW gap (pp)
20 epochs
50 epochs
0
5
10
15
20
25
↓2.3pp
+9.0pp
+6.8pp
0pp
+10.3pp
+10.4pp
↓3.5pp
+16.1pp
+12.6pp
↓7.9pp
+11.5pp
+3.6pp
↓12.9pp
+24.3pp
+11.4pp
C100
ResNet-50
C100
DeiT-S
C100
ConvNeXt-T
IN100
ResNet-50
IN100
ConvNeXt-T